

AI Based Mock Interview System

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Abstract: In the current Job market , Students and job seekers often struggle to access realistic interview practice and structured feedback, which are essential for improving performance. Traditional methods like reading questions or limited mock sessions fail to provide real interview exposure or systematic evaluation. To address this, the project proposes an AI-Based Mock Interview System, a web platform that simulates role-based interviews. Users can respond through text or audio, with optional camera access for a realistic experience. Responses are converted to text and evaluated using keyword matching and similarity analysis against predefined ideal answers. The system provides structured feedback highlighting strengths and areas for improvement, while also storing interview history to track progress. It focuses on explainable, practical evaluation methods rather than complex analysis, ensuring reliability. The expected outcome is improved interview readiness through consistent practice, evaluation, and actionable feedback, integrating web technologies, data analysis, and applied AI to solve a real-world problem.

Keywords: Mock Interview System, Interview Simulation, Answer Evaluation, Artificial Intelligence, Web Application, Interview Preparation

Introduction: The Mock Interview System is an AI-powered platform designed to provide a realistic and interactive interview experience for job seekers. Built with a React frontend and a Python-Flask backend, the system leverages advanced algorithms for real-time face detection, emotion analysis, and cheat detection to ensure interview integrity. Candidates can answer questions through integrated speech recognition, mimicking the flow of a real conversation. Following the session, the platform generates comprehensive performance reports, offering personalized feedback based on technical accuracy and behavioral cues. This holistic approach helps users refine their interviewing skills and build confidence in a simulated environment

Motivation: The primary motivation for this project is to address the gap in interview preparation by providing a scalable, AI-driven solution for job seekers to practice in a realistic environment. By offering objective feedback on technical performance, emotional cues, and communication, the system helps users overcome interview anxiety and build the confidence necessary to succeed in their professional careers.

Methodology: The methodology integrates a React-based frontend for seamless user interaction with a Python-Flask backend that processes AI-driven logic. It employs advanced computer vision algorithms for real-time face and emotion analysis, alongside integrated speech recognition to evaluate candidate responses and deliver data-driven performance reports.

Result analysis: The success of the system is measured by its ability to accurately detect candidate emotions and potential cheating indicators while providing high-quality, AI-generated performance reviews. Users documented increased confidence and improved communication skills after multiple sessions, indicating the platform's effectiveness as a reliable preparation tool.

Conclusion: The AI Mock Interview System provides a robust and interactive platform for job seekers to refine their interviewing skills through a combination of technical evaluation and behavioral analysis. By integrating real-time face detection and speech recognition, the system ensures a comprehensive assessment that goes beyond simple automated questioning. The feedback generated after each session offers actionable insights, helping candidates identify their strengths and areas for improvement in a realistic simulation. This innovative tool not only enhances technical preparation but also builds the confidence needed to succeed in competitive professional environments.

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